

```
import pandas as pd
```

```
import numpy as np
```

```
auto_price_df = pd.read_csv('/content/sample_data/Automobile_price_data__Raw (1).csv')
```

```
auto_price_df.info()
```

```
↗ <class 'pandas.core.frame.DataFrame'>  
RangeIndex: 205 entries, 0 to 204  
Data columns (total 26 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   symboling              205 non-null    int64  
1   normalized-losses      205 non-null    object  
2   make                   205 non-null    object  
3   fuel-type              205 non-null    object  
4   aspiration              205 non-null    object  
5   num-of-doors            205 non-null    object  
6   body-style              205 non-null    object  
7   drive-wheels            205 non-null    object  
8   engine-location         205 non-null    object  
9   wheel-base              205 non-null    float64  
10  length                  205 non-null    float64  
11  width                   205 non-null    float64  
12  height                  205 non-null    float64  
13  curb-weight             205 non-null    int64  
14  engine-type             205 non-null    object  
15  num-of-cylinders        205 non-null    object  
16  engine-size             205 non-null    int64  
17  fuel-system             205 non-null    object  
18  bore                    205 non-null    object  
19  stroke                  205 non-null    object  
20  compression-ratio       205 non-null    float64  
21  horsepower              205 non-null    object  
22  peak-rpm                205 non-null    object  
23  city-mpg                205 non-null    int64  
24  highway-mpg             205 non-null    int64  
25  price                   205 non-null    object  
dtypes: float64(5), int64(5), object(16)  
memory usage: 41.8+ KB
```

```
auto_price_df.replace("?", np.nan, inplace = True)
```

```
auto_price_df
```



	symboling	normalized- losses	make	fuel- type	aspiration	num- of- doors	body- style	drive- wheels	engine locatic
0	3	NaN	alfa-romero	gas	std	two	convertible	rwd	fro
1	3	NaN	alfa-romero	gas	std	two	convertible	rwd	fro
2	1	NaN	alfa-romero	gas	std	two	hatchback	rwd	fro
3	2	164	audi	gas	std	four	sedan	fwd	fro
4	2	164	audi	gas	std	four	sedan	4wd	fro
...	...	...	...	...	...	...	...	...	
200	-1	95	volvo	gas	std	four	sedan	rwd	fro
201	-1	95	volvo	gas	turbo	four	sedan	rwd	fro
202	-1	95	volvo	gas	std	four	sedan	rwd	fro
203	-1	95	volvo	diesel	turbo	four	sedan	rwd	fro
204	-1	95	volvo	gas	turbo	four	sedan	rwd	fro

205 rows × 26 columns

```
auto_price_df.isnull().sum()
```



	0
symboling	0
normalized-losses	41
make	0
fuel-type	0
aspiration	0
num-of-doors	2
body-style	0
drive-wheels	0
engine-location	0
wheel-base	0
length	0
width	0
height	0
curb-weight	0
engine-type	0
num-of-cylinders	0
engine-size	0
fuel-system	0
bore	4
stroke	4
compression-ratio	0
horsepower	2
peak-rpm	2
city-mpg	0
highway-mpg	0
price	4

dtype: int64

auto_price_df['make']

**make**

0	alfa-romero
1	alfa-romero
2	alfa-romero
3	audi
4	audi
...	...
200	volvo
201	volvo
202	volvo
203	volvo
204	volvo

205 rows × 1 columns

dtype: object

```
auto_price_df['make'].value_counts()
```



	count
make	
toyota	32
nissan	18
mazda	17
mitsubishi	13
honda	13
volkswagen	12
subaru	12
peugot	11
volvo	11
dodge	9
mercedes-benz	8
bmw	8
audi	7
plymouth	7
saab	6
porsche	5
isuzu	4
jaguar	3
chevrolet	3
alfa-romero	3
renault	2
mercury	1

dtype: int64

```
auto_price_df['make'] == 'audi'
```

**make****0** False**1** False**2** False**3** True**4** True

... ..

200 False**201** False**202** False**203** False**204** False

205 rows × 1 columns

dtype: bool

auto_price_df[auto_price_df['make'] == 'audi']



	symboling	normalized- losses	make	fuel- type	aspiration	num- of- doors	body- style	drive- wheels	engine- location	w
3	2	164	audi	gas	std	four	sedan	fwd	front	
4	2	164	audi	gas	std	four	sedan	4wd	front	
5	2	NaN	audi	gas	std	two	sedan	fwd	front	
6	1	158	audi	gas	std	four	sedan	fwd	front	
7	1	NaN	audi	gas	std	four	wagon	fwd	front	
8	1	158	audi	gas	turbo	four	sedan	fwd	front	
9	0	NaN	audi	gas	turbo	two	hatchback	4wd	front	

7 rows × 26 columns

auto_price_df[auto_price_df['highway-mpg'] >= 40]



	symboling	normalized- losses	make	fuel- type	aspiration	num- of- doors	body- style	drive- wheels	eng loc:
18	2	121	chevrolet	gas	std	two	hatchback	fwd	
19	1	98	chevrolet	gas	std	two	hatchback	fwd	
20	0	81	chevrolet	gas	std	four	sedan	fwd	
21	1	118	dodge	gas	std	two	hatchback	fwd	
30	2	137	honda	gas	std	two	hatchback	fwd	
32	1	101	honda	gas	std	two	hatchback	fwd	
44	1	NaN	isuzu	gas	std	two	sedan	fwd	
45	0	NaN	isuzu	gas	std	four	sedan	fwd	
63	0	NaN	mazda	diesel	std	NaN	sedan	fwd	
76	2	161	mitsubishi	gas	std	two	hatchback	fwd	
90	1	128	nissan	diesel	std	two	sedan	fwd	
118	1	119	plymouth	gas	std	two	hatchback	fwd	
159	0	91	toyota	diesel	std	four	hatchback	fwd	
160	0	91	toyota	gas	std	four	sedan	fwd	
182	2	122	volkswagen	diesel	std	two	sedan	fwd	
184	2	94	volkswagen	diesel	std	four	sedan	fwd	
187	2	94	volkswagen	diesel	turbo	four	sedan	fwd	

17 rows × 26 columns

```
auto_price_df[(auto_price_df['highway-mpg'] >= 40) & (auto_price_df['body-style'] == 'sedan']
```



	symboling	normalized-losses	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine locatic
20	0	81	chevrolet	gas	std	four	sedan	fwd	fro
44	1	NaN	isuzu	gas	std	two	sedan	fwd	fro
45	0	NaN	isuzu	gas	std	four	sedan	fwd	fro